

- 3 (a) *either* energy (stored)/work done represented by area under graph B1
or energy = average force $\times$ extension C1
energy = $\frac{1}{2} \times 180 \times 4.0 \times 10^{-2}$ A1 [3]
= 3.6 J
- (b) (i) *either* momentum before release is zero M1
so sum of momenta (of trolleys) after release is zero A1
or force = rate of change of momentum (M1)
force on trolleys equal and opposite (A1)
or impulse = change in momentum (M1)
impulse on each equal and opposite (A1) [2]
- (ii) 1 $M_1 V_1 = M_2 V_2$ B1 [1]
2 $E = \frac{1}{2} M_1 V_1^2 + \frac{1}{2} M_2 V_2^2$ B1 [1]
- (iii) 1 $E_K = \frac{1}{2} m v^2$ and $p = m v$ combined to give M1
 $E_K = p^2 / 2m$ A0 [1]
2 m smaller, E_K is larger because p is the same/constant M1
so trolley B A0 [1]

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4 (a) when a wave (front) passes by/incident on an edge/corner

M1